

AI / ML INTERNSHIP PROGRAM DEVIXO SOLUTIONS

Task 01 Python Programming, Data Analysis & Visualization

Version 1.0 approved

Prepared by <Muhammad Hamza Waseem>

Submission Date : 13-09-26

1. Introduction	1
2. Objective	2
3. Dataset Name	2
4. Dataset Source	2
5. Dataset Description	2
6. Dataset Size	2
7. Features/Columns Description	2
8. Methodology	4
9. Data Loading	4
10. Data Cleaning	4
11. Exploratory Data Analysis (EDA)	4
12. Statistical Analysis	4
13. Data Visualizations	4
14. Code Screenshots	4
15. Output Screenshots	6
16. Graph Screenshots	7
17. Major Findings	9
18. At Least 10 Insights	9
19. Conclusion	10
20. GitHub Repository Link	10
21. Streamlit Dashboard Screenshots	10

1. Introduction

Understanding why some students get high grades while others struggle is important for improving education. This project looks closely at student habits, school attendance, and study hours to see what really drives academic success.

2. Objective

The main goal is to find out which factors impact exam scores the most, clean and study the data using Python, and build a simple interactive web dashboard to show the results clearly.

3. Dataset Name

Student Performance Factors Dataset (`StudentPerformanceFactors.csv`).

4. Dataset Source

A public data repository containing student lifestyle, study habits, and academic records.

5. Dataset Description

This dataset includes details about how many hours students study, their attendance rates, tutoring sessions, parental support, previous scores, and their final exam scores.

6. Dataset Size

The dataset consists of multiple rows (representing individual student records) and columns (representing different features).

7. Features/Columns Description

- ❖ `Hours_Studied`: Total number of hours each student spent studying.
- ❖ `Attendance`: Percentage of classes attended by the student.
- ❖ `Parental_Involvement`: Level of family support and guidance provided to the student (e.g., Low, Medium, High).
- ❖ `Access_to_Resources`: Availability of learning tools like textbooks, computers, and internet access.
- ❖ `Extracurricular_Activities`: Participation in sports, clubs, or other activities outside of regular classes.

- ❖ Sleep_Hours: Average number of hours the student sleeps each night.
- ❖ Previous_Scores: Marks obtained by the student in earlier exams.
- ❖ Motivation_Level: The student's inner drive and interest in learning (e.g., Low, Medium, High).
- ❖ Internet_Access: Whether the student has reliable internet connection at home (Yes or No).
- ❖ Tutoring_Sessions: Number of extra tutoring or coaching classes attended per month.
- ❖ Family_Income: Economic background level of the student's family (e.g., Low, Medium, High).
- ❖ Teacher_Quality: Rating of teaching standards experienced by the student.
- ❖ School_Type: The category of school the student attends (e.g., Public or Private).
- ❖ Peer_Influence: The impact of peer groups on the student's academic behavior (e.g., Positive, Neutral, Negative).
- ❖ Physical_Activity: Average hours spent weekly on physical exercise or sports.
- ❖ Learning_Disabilities: Presence of any learning challenges or conditions (Yes or No).
- ❖ Parental_Education_Level: Highest education level completed by the parents.
- ❖ Distance_from_Home: How far the student lives from the school.
- ❖ Gender: The gender of the student.
- ❖ Exam_Score: Final marks obtained by the student in the final examination.

8. Methodology

The project follows a step-by-step approach: loading the data with Pandas, cleaning missing values, running statistical summaries, building 10 different visualizations using Matplotlib and Seaborn, and launching a Streamlit dashboard.

9. Data Loading

Python's Pandas library was used to load the CSV file into a DataFrame using `pd.read_csv("StudentPerformanceFactors.csv")`.

10. Data Cleaning

The dataset was checked for missing values, duplicates, and incorrect data types to ensure the analysis results are accurate and reliable.

11. Exploratory Data Analysis (EDA)

Exploratory analysis was performed to check data distributions, find general averages, and spot basic trends and patterns among student variables.

12. Statistical Analysis

Descriptive statistics (`df.describe()`) were calculated to find the mean, minimum, maximum, and spread of numerical columns like exam scores and study hours.

13. Data Visualizations

Ten distinct visualizations were created using Matplotlib and Seaborn, including scatter plots, heatmaps, histograms, box plots, and a violin plot.

14. Code Screenshots

PART 1 – DATA LOADING & DATA UNDERSTANDING

```
1 import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

LOADING DATASET

```
1 df=pd.read_csv("StudentPerformanceFactors.csv")
```

DISPLAYING FIRST 5 RECORDS

Number of missing values in columns

```
▶ df.isnull().sum()
```

```
▶ df['Parental_Education_Level'].unique()
mode_value=df["Parental_Education_Level"].mode()[0]
df["Parental_Education_Level"]=df["Parental_Education_Level"].fillna(mode_value)
```

All categorical columns missing values were filled with the most frequent category (Mode) to preserve the original data distribution.

```
df['Teacher_Quality'].unique()
mode_value = df["Teacher_Quality"].mode()[0]
df["Teacher_Quality"]=df["Teacher_Quality"].fillna(mode_value)
```

Checking numerical cols

```
num_cols=df.select_dtypes(include="number").columns
print(num_cols)

Index(['Hours_Studied', 'Attendance', 'Sleep_Hours', 'Previous_Scores',
      'Tutoring_Sessions', 'Physical_Activity', 'Exam_Score'],
      dtype='object')
```

Checking categorical or object cols

```
categorical_cols=df.select_dtypes(include=["object", "category"]).columns
print(categorical_cols)

Index(['Parental_Involvement', 'Access_to_Resources',
      'Extracurricular_Activities', 'Motivation_Level', 'Internet_Access',
      'Family_Income', 'Teacher_Quality', 'School_Type', 'Peer_Influence',
      'Learning_Disabilities', 'Parental_Education_Level',
      'Distance_from_Home', 'Gender'],
      dtype='object')
```

Checking datetime cols

```
datetime_cols = df.select_dtypes(include=['datetime64', 'datetimez']).columns
print(datetime_cols)

Index([], dtype='object')
```

15. Output Screenshots

me	Teacher_Quality	School_Type	Peer_Influence	Physical_Activity	Learning_Disabilities	Parental_Education_Level	Distance_from_Home	Gender	Exam_Score
ow	Medium	Public	Positive	3	No	High School	Near	Male	67
jm	Medium	Public	Negative	4	No	College	Moderate	Female	61
jm	Medium	Public	Neutral	4	No	Postgraduate	Near	Male	74
jm	Medium	Public	Negative	4	No	High School	Moderate	Male	71
jm	High	Public	Neutral	4	No	College	Near	Female	70

```
Index(['Hours_Studied', 'Attendance', 'Parental_Involvement',
      'Access_to_Resources', 'Extracurricular_Activities', 'Sleep_Hours',
      'Previous_Scores', 'Motivation_Level', 'Internet_Access',
      'Tutoring_Sessions', 'Family_Income', 'Teacher_Quality', 'School_Type',
      'Peer_Influence', 'Physical_Activity', 'Learning_Disabilities',
      'Parental_Education_Level', 'Distance_from_Home', 'Gender',
      'Exam_Score'],
      dtype='object')
```

```

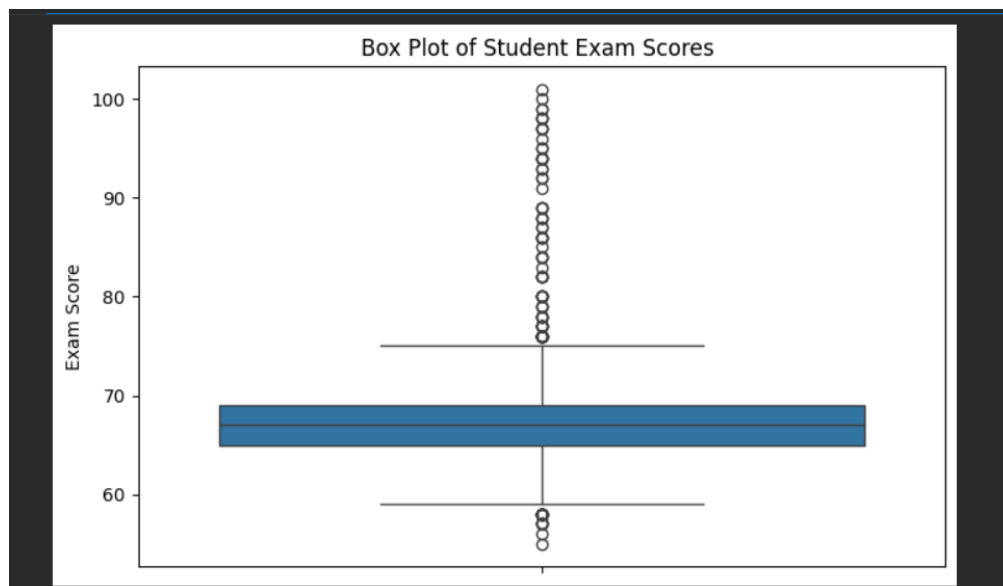
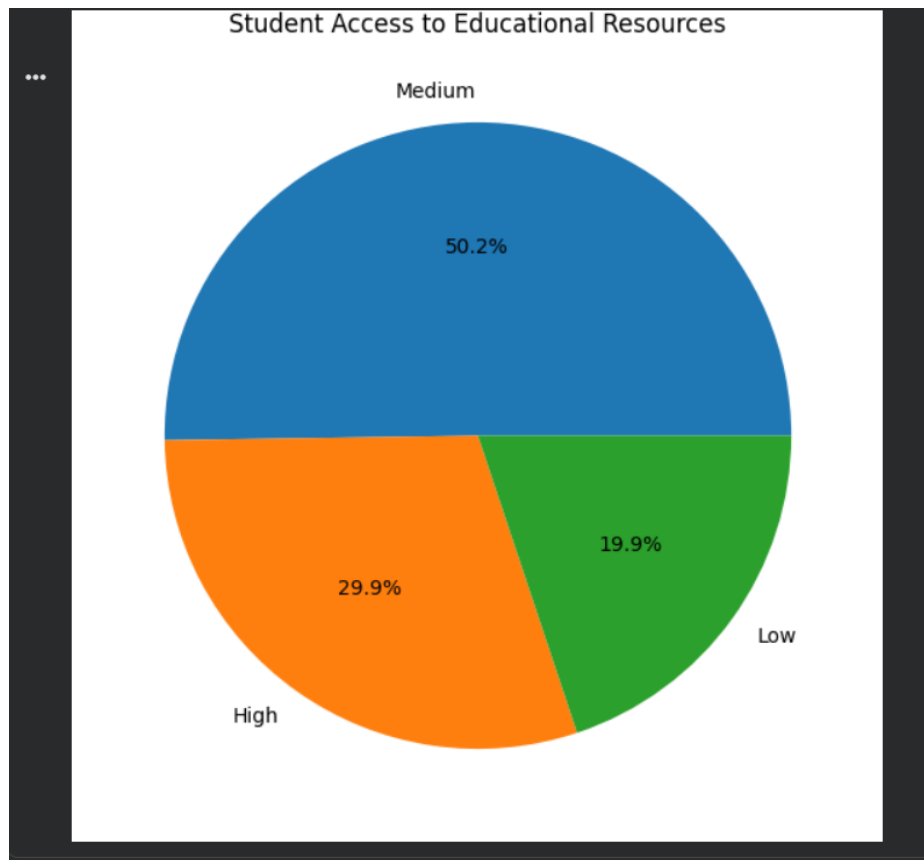
... <class 'pandas.core.frame.DataFrame'>
RangeIndex: 6607 entries, 0 to 6606
Data columns (total 20 columns):
 #   Column                                Non-Null Count  Dtype
---  -
 0   Hours_Studied                        6607 non-null   int64
 1   Attendance                          6607 non-null   int64
 2   Parental_Involvement                6607 non-null   object
 3   Access_to_Resources                 6607 non-null   object
 4   Extracurricular_Activities          6607 non-null   object
 5   Sleep_Hours                        6607 non-null   int64
 6   Previous_Scores                     6607 non-null   int64
 7   Motivation_Level                    6607 non-null   object
 8   Internet_Access                     6607 non-null   object
 9   Tutoring_Sessions                   6607 non-null   int64
10   Family_Income                       6607 non-null   object
11   Teacher_Quality                     6529 non-null   object
12   School_Type                         6607 non-null   object
13   Peer_Influence                      6607 non-null   object
14   Physical_Activity                   6607 non-null   int64
15   Learning_Disabilities               6607 non-null   object
16   Parental_Education_Level            6517 non-null   object
17   Distance_from_Home                  6540 non-null   object
18   Gender                             6607 non-null   object
19   Exam_Score                          6607 non-null   int64
dtypes: int64(7), object(13)
memory usage: 1.0+ MB

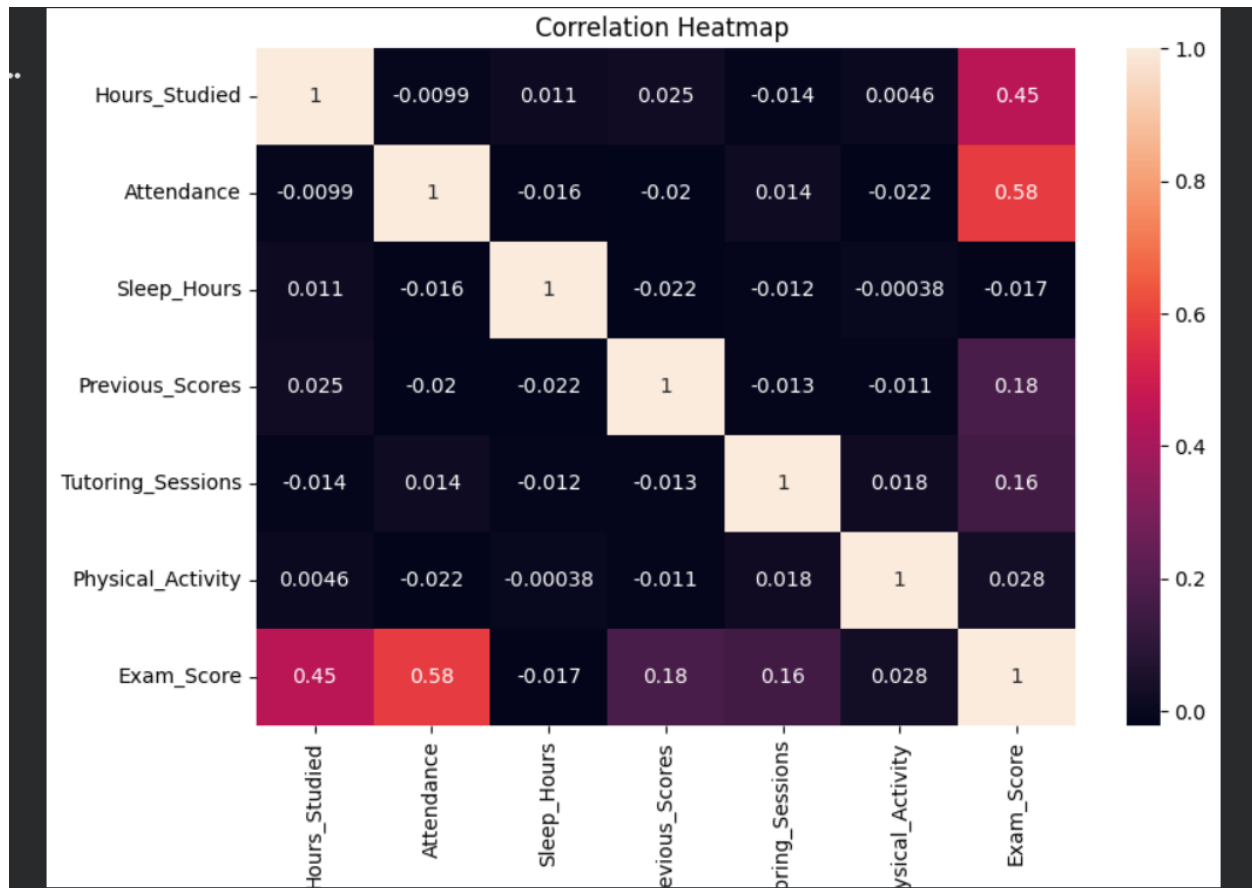
```

...

	0
Hours_Studied	0
Attendance	0
Parental_Involvement	0
Access_to_Resources	0
Extracurricular_Activities	0
Sleep_Hours	0
Previous_Scores	0
Motivation_Level	0
Internet_Access	0
Tutoring_Sessions	0
Family_Income	0
Teacher_Quality	78
School_Type	0
Peer_Influence	0
Physical_Activity	0
Learning_Disabilities	0
Parental_Education_Level	90
Distance_from_Home	67

16. Graph Screenshots





17. Major Findings

Attendance and hours studied are the two biggest positive drivers of final exam scores. In contrast, auxiliary lifestyle habits like sleep and physical activity showed very little direct impact on grades in this cohort.

18. At Least 10 Insights

- **Main positive link:** Attendance has the strongest connection with Exam_Score, meaning students who attend class regularly get better grades.
- **Study time impact:** Hours_Studied also has a positive link, meaning studying more hours clearly helps students score higher marks.
- **Sleep and fitness:** Factors like Sleep_Hours and Physical_Activity show almost no direct impact on exam scores.
- **Score spread:** Most student scores are in the middle range, with only a few students getting very low or very high grades.

- **Performance groups:** The student count shows that the majority fall into the medium performance group.
- **Parental support:** Higher family guidance and support lead to better average exam scores.
- **Unusual scores:** The box plot highlights a few students with unusually high or low scores compared to the rest.
- **Overall trends:** Pair plots show that past scores, study hours, and attendance all connect well with final exam results.
- **Tutoring classes:** The tutoring graph shows how extra help sessions shift score groups.
- **Final takeaway:** To help students get better grades, schools should focus mostly on improving regular attendance and daily study habits.

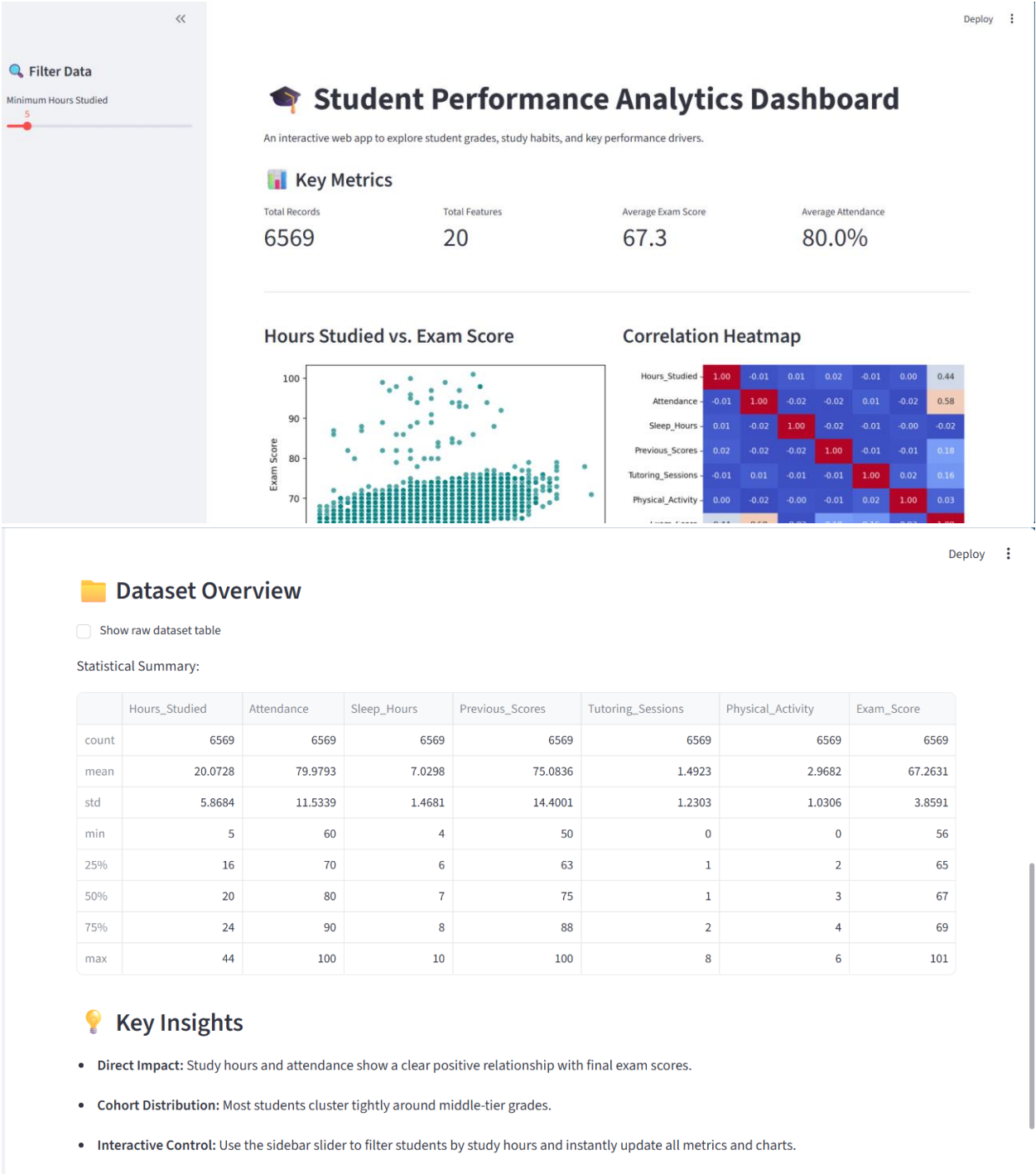
19. Conclusion

This project successfully analyzed student performance data to separate hard facts from assumptions. By cleaning the data, running statistical checks, and building ten visualizations plus a Streamlit dashboard, we proved that regular attendance and study hours are essential for high grades. Schools can use these insights to build better support systems for struggling students.

20. GitHub Repository Link

<https://github.com/HamzaWaseem2005/AI-ML>

21. Streamlit Dashboard Screenshots



Key Insights

- Direct Impact:** Study hours and attendance show a clear positive relationship with final exam scores.
- Cohort Distribution:** Most students cluster tightly around middle-tier grades.
- Interactive Control:** Use the sidebar slider to filter students by study hours and instantly update all metrics and charts.